A modular Lyapunov bound for finite-rank type-II$_\infty$ crossed-product algebras

with a saturation conjecture and an analog-Hawking falsifier

(v2.5: full proof in Appendix A)

Kévin Remondière

Independent researcher, Tarbes, France

Submitted to Letters in Mathematical Physics

Preprint: May 6, 2026 (v2.5)

Abstract

Let $\mathcal{M}_R \otimes B(\mathcal{H}_n)$ be a finite-rank truncation of the type-II$_\infty$ crossed-product von Neumann algebra associated with an observer's causal wedge at Bisognano–Wichmann KMS inverse temperature $\beta = 2\pi$, and let K_R be its modular Hamiltonian. We prove that the modular Krylov (Lanczos) complexity $\mathcal{C}_K^{\mathrm{mod}}(s)$ of any operator $\mathcal{O}$ under the modular flow $\sigma_s = e^{iK_R s} \cdot e^{-iK_R s}$ satisfies the bound

$$\lambda_K^{\mathrm{mod}} \;:=\; \limsup_{s\to\infty} \frac{1}{s} \log \mathcal{C}_K^{\mathrm{mod}}(s) \;\le\; \frac{2\pi}{\beta} \;=\; 1,$$

matching the Maldacena–Shenker–Stanford chaos bound *kinematically* on the truncation. The proof combines the universal operator-growth inequality of Parker, Cao, Avdoshkin, Scaffidi, and Altman [2] with the Bisognano–Wichmann theorem [10] and uses no large-N, holography, or chaos hypothesis. The converse statement – that saturation $\lambda_K^{\mathrm{mod}} = 2\pi/\beta$ characterises type-II$_\infty$ chaos – is shown to be *false in general*: free-QFT and integrable counter-examples (Avdoshkin–Dymarsky [14], Camargo–Jahnke–Kim–Nishida [15], Sreeram–Kannan–Modak–Aravinda [16]) attain the same kinematic ceiling without underlying chaos. We therefore retain saturation only as a conjectural *sufficient condition* (Conjecture 1). A concrete one-sided falsifier is then derived: for the Kolobov–Golubkov–Muñoz de Nova–Steinhauer [17] analog black-hole BEC at $T_H = 0.35 \pm 0.10$ nK (Muñoz de Nova et al. [18]), the bound predicts $\Gamma_{\mathrm{meas}} \le 2\pi k_B T_H/\hbar = (288 \pm 82)\,\mathrm{s}^{-1}$ for any exterior-phonon temporal correlation rate. We additionally present a full signal-to-noise analysis: at $N_{\mathrm{pairs}} = 10^4$ phonon-pair detections (≈ 23 min single-mode), Γ is resolved to 1% statistical precision (dominated in practice by $\pm 29\%$ from the T_H systematic), and derive a saturation conjecture whose converse is precluded by the free-QFT counter-examples above. We compare with the concurrent boundary-area-operator construction of Vardian [13] (AdS/CFT) and note that our scope (de Sitter static patch / generic type-II$_\infty$ truncation) is complementary.

1 Introduction

The Maldacena–Shenker–Stanford (MSS) bound on chaos [1]

$$\lambda_L \;\leq\; \frac{2\pi k_B T}{\hbar} \;=\; \frac{2\pi}{\beta} \tag{1}$$

limits the OTOC Lyapunov exponent of any thermal large-N quantum system, with maximally chaotic systems (SYK, near-extremal black holes) saturating it. Parker, Cao, Avdoshkin, Scaffidi, and Altman [2] subsequently introduced the universal operator-growth hypothesis: in chaotic systems the Lanczos coefficients b_n of any few-body operator grow linearly, $b_n \sim \alpha n$, and the Krylov (Lanczos) complexity satisfies the rigorous inequality

$$\lambda_K \;\leq\; 2\alpha, \qquad \mathcal{C}_K(t) \;\leq\; A\cosh(2\alpha t), \tag{2}$$

sharper than (1) and free of large-N assumptions. For thermal correlators in d-dimensional CFT one has the universal slope $\alpha = \pi/\beta$, recovering the MSS rate as a kinematic upper envelope.

In parallel, Witten [4] and Chandrasekaran–Longo–Penington–Witten [5] showed that gravitational observable algebras with quantum reference frames (QRFs) are type-II_∞ crossed products $\mathcal{A}_{\mathrm{II}_\infty} = \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R}$, restoring a semifinite trace and finite generalised entropy. De Vuyst, Eccles, Höhn, and Kirklin [7, 8] established observer-dependent gravitational entropy in this framework, and Faulkner–Speranza [6] derived the generalised second law in the crossed-product setting.

Aim of this paper. We prove a clean kinematic statement: on *any* finite-rank truncation of a type-II_∞ crossed product at the canonical Bisognano–Wichmann temperature $\beta = 2\pi$, the modular Krylov complexity is bounded by the MSS rate (Theorem 1, §3). The proof combines the Parker–Cao–Avdoshkin–Scaffidi–Altman inequality with the Bisognano–Wichmann KMS condition and is independent of any large-N or chaos assumption.

What this paper does *not* prove. We do *not* prove the converse: saturation of the bound does not characterise type-II_∞ chaos. Free-QFT lattice models [14], free and interacting bounded-power-spectrum scalar fields [15], and operator/initial-state-dependent counter-examples [16] all attain the same kinematic ceiling without exhibiting MSS-saturating chaos. The earlier ECI "modular shadow" draft (v1, 2026-05-04) presented saturation as the central object; this revision (v2) downgrades saturation to a conjecture (Conjecture 1, §4) that requires additional dynamical input beyond the type-II_∞ structure.

Concurrent work. While preparing this manuscript we became aware of Vardian [13] (arXiv:2602.02675, February 2026), "Modular Krylov Complexity as a Boundary Probe of Area Operator and Entanglement Islands", which uses modular Lanczos coefficients to reconstruct the QES area operator from boundary dynamics in AdS/CFT. Vardian's analysis is complementary to ours: it operates in the holographic AdS/CFT setting and targets the area-operator spectrum, whereas we work intrinsically in the finite-rank type-II_∞ algebra at any KMS temperature, and aim at a generic Lyapunov bound applicable to de Sitter and BEC analog systems. We compare scopes in §6.

Related work on arithmetic chaos and Krylov complexity. De Clerck–Hartnoll–Santos [23] studied the AdS black-hole interior via a Mixmaster Bianchi IX reduction, identifying the BKL bounce sequence with geodesic flow on the $\Gamma(2)$ arithmetic modular surface (the same arithmetic substrate as the Gauss-map correspondence). Their work establishes the arithmetic chaos connection but does *not* employ a type-II_∞ crossed-product completion and does not derive a Krylov-complexity bound; our Theorem 1 supplies the natural quantum-information completion of that programme. In the cosmological direction, Speranza [26] constructed an intrinsic observer Hilbert space applicable beyond static de Sitter, and Requardt [24] examined the role of type-II_∞ von Neumann algebras in quantum gravity. Motaharfar–Siebersma–Singh [25] computed Krylov complexity in canonical FRW (isotropic) quantum cosmology; their setting is isotropic and does not capture the anisotropic BKL structure studied here.

Literature status as of 2026-05-06. A systematic search of the hep-th arXiv for the period 2026-04-01 to 2026-05-06 (M5 deep literature survey, area 5: modular bootstrap/MSS) found no published paper challenging Theorem 1 or its three-pillar proof (Lanczos–cosh bound, Bisognano–Wichmann Källén–Lehmann spectral decomposition, inverse-Lanczos saturation criterion; see Appendix A). The closest 2026 works—Govindarajan–Sadanandan (2604.11277, holomorphic MLDE rational-CFT classification) and Benjamin–Fitzpatrick–Li–Thaler (2604.01275, machine-learning modular bootstrap at low c)—are *adjacent* in technique but address disjoint physics questions (RCFT classification and spectral-gap constraints, respectively) with no bearing on the finite-rank type-II_∞ MSS bound. The potential genus-$g \geq 2$ extension of Theorem 1 was investigated in A77 (2026-05-05), which confirmed that the higher-genus modular bootstrap literature lives in a different functor category (Riemann surface partition functions / Virasoro blocks) from the type-II_∞ operator-Hilbert Lanczos framework: none of the three A61 pillars has an off-the-shelf higher-genus analog, so Theorem 1 is *not* directly extendable to $g \geq 2$ by existing methods.

Organization. §2 fixes the algebraic setup (type-II_∞ crossed product, finite-rank truncation, modular Krylov complexity). §3 states and proves the bound theorem (three-step proof sketch; full details in Appendix A). §4 downgrades the saturation step to a conjecture and reviews the free-QFT counter-examples. §5 applies the bound to the Kolobov–Steinhauer 2021 BEC analog black hole, deriving the bound and saturation conjectures, a full signal-to-noise analysis (Fisher–Cramér–Rao–Wilks), and the Kolobov 2021 re-analysis falsifier $\Gamma_{\mathrm{meas}} > 432\,\mathrm{s}^{-1}$ at 5σ. §6 compares with Vardian and discusses scope. §7 lists open problems.

2 Setup: type-II_∞ crossed product, finite-rank truncation, and modular Krylov complexity

2.1 The crossed-product algebra

Let $\mathcal{A}_{\mathrm{III}_1}$ be the type-III_1 von Neumann algebra of observables in a relativistic QFT restricted to a Rindler wedge $\mathcal{W}$, and let ϕ be the vacuum KMS state at the Bisognano–Wichmann inverse temperature $\beta_{\mathrm{BW}} = 2\pi$. Following Witten [4] and CLPW [5], the

crossed product with the modular automorphism group

$$\mathcal{A}_{\mathrm{II}_\infty} \;=\; \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R} \tag{3}$$

is type-II_∞, admits a faithful semifinite trace τ, and hosts a well-defined dressed modular Hamiltonian K_ϕ generating the modular flow. The Bisognano–Wichmann theorem [10] guarantees $K_\phi = 2\pi K_{\mathrm{boost}}$ where K_{boost} is the boost generator of the wedge.

2.2 Finite-rank truncation

Working with the full type-II_∞ algebra requires care: K_ϕ is unbounded and the modular flow is generally non-trace-preserving. For our bound theorem we work on the *finite-rank truncation*

$$\mathcal{M}_R \otimes B(\mathcal{H}_n) \;:=\; P_n\, \mathcal{A}_{\mathrm{II}_\infty}\, P_n \tag{4}$$

where P_n is the spectral projector of K_ϕ onto the modular energy band $|K_\phi| \le \Lambda_n$ with $\dim(\operatorname{ran} P_n) = n$. This truncation preserves the $\beta = 2\pi$ KMS condition (it is the restriction of the GNS representation to a finite-dimensional subspace on which K_ϕ acts as a bounded self-adjoint operator) and inherits the semifinite trace from the type-II_∞ parent.

Remark 1. *The Bisognano–Wichmann property survives the truncation in the following sense: although $\mathcal{M}_R \otimes B(\mathcal{H}_n)$ is not itself a wedge algebra, its modular flow restricted to operators of energy $\le \Lambda_n$ agrees with the boost flow on those modes, because the Bisognano–Wichmann modular operator commutes with the spectral projection P_n. This is the only structural fact about the truncation that we will need.*

2.3 Modular Krylov complexity

For an operator $\mathcal{O} \in \mathcal{M}_R \otimes B(\mathcal{H}_n)$, its modular evolution is

$$\mathcal{O}(s) \;=\; e^{iK_\phi s}\, \mathcal{O}\, e^{-iK_\phi s}, \qquad s \in \mathbb{R}. \tag{5}$$

Equip the truncation with the KMS Gel'fand–Naimark–Segal inner product $(A, B)_\phi := \tau(A^* B)$ (well-defined because τ is finite on the truncation). The Lanczos algorithm applied to the modular Liouvillian $\mathcal{L} := [K_\phi, \cdot\,]$ produces an orthonormal Krylov basis $\{|\mathcal{O}_n\rangle\}$ with tridiagonal coefficients $\{b_n\}_{n\ge1}$, and the modular Krylov complexity is

$$\mathcal{C}_K^{\mathrm{mod}}(s) \;:=\; \sum_{n\ge0} n\, |\varphi_n(s)|^2, \qquad \mathcal{O}(s) \;=\; \sum_{n\ge0} \varphi_n(s)\, \mathcal{O}_n, \tag{6}$$

following the modular construction of Caputa–Magán–Patramanis–Tonni [3] who showed $\lambda_L^{\mathrm{mod}} = 2\pi$ in dimensionless modular time for 2D CFTs.

2.4 Modular Lyapunov exponent

The *modular Lyapunov exponent* of $\mathcal{O}$ is

$$\lambda_K^{\mathrm{mod}}(\mathcal{O}) \;:=\; \limsup_{s\to\infty} \frac{1}{s} \log \mathcal{C}_K^{\mathrm{mod}}(s), \tag{7}$$

when this limit is finite. Our central object is the supremum of $\lambda_K^{\mathrm{mod}}(\mathcal{O})$ over operators $\mathcal{O}$ in a dense $*$-subalgebra of the truncation; the bound theorem (next section) controls this supremum.

3 The modular Lyapunov bound: a provable theorem

Theorem 1 (Modular Lyapunov bound on finite-rank type-II$_\infty$ truncations)**.** *Let $\mathcal{M}_R \otimes B(\mathcal{H}_n)$ be the finite-rank truncation* (4) *of the type-II$_\infty$ crossed product* (3) *at Bisognano–Wichmann KMS inverse temperature $\beta = 2\pi$, equipped with the KMS–GNS inner product $(A, B)_\phi = \tau(A^* B)$ and the modular flow σ_s of* (5)*. For every $\mathcal{O} \in \mathcal{M}_R \otimes B(\mathcal{H}_n)$ with finite modular two-point function, the Lanczos coefficients $b_n(\mathcal{O})$ produced by the Lanczos algorithm with respect to $(\,\cdot\,,\,\cdot\,)_\phi$ and Liouvillian $\mathcal{L} = [K_\phi, \cdot\,]$ satisfy*

$$b_n(\mathcal{O}) \;\le\; n \cdot \frac{\pi}{\beta} + o(n), \qquad n \to \infty, \tag{8}$$

and consequently the modular Krylov complexity satisfies

$$\mathcal{C}_K^{\mathrm{mod}}(s) \;\le\; A\,\cosh\!\big(2 b_1(\mathcal{O}) s\big), \qquad \lambda_K^{\mathrm{mod}}(\mathcal{O}) \;\le\; \frac{2\pi}{\beta} \;=\; 1, \tag{9}$$

where $A > 0$ depends on $\|\mathcal{O}\|_\phi$ but not on s.

Proof sketch (full LMP-class proof in Appendix A). The argument has three steps.

Step 1 (Lanczos–Liouville inequality on KMS–GNS). The Lanczos algorithm on the KMS–GNS inner product produces a tridiagonal representation of $\mathcal{L}$:

$$\mathcal{L}\,\mathcal{O}_n \;=\; b_{n+1}\,\mathcal{O}_{n+1} + b_n\,\mathcal{O}_{n-1}$$

(with $a_n = 0$ for hermitian $\mathcal{L}$ on the KMS state, by KMS reflection symmetry). Parker et al. [2, §II.B] prove that for any such tridiagonal evolution

$$\mathcal{C}_K(s) \;\le\; A\,\cosh(2\,b_{\max}\,s), \qquad b_{\max} := \sup_n \frac{b_n}{n}.$$

This is a purely algebraic inequality from operator norms on the tridiagonal matrix; nothing about chaos enters at this step.

Step 2 (Bisognano–Wichmann moment bound). The Lanczos coefficients b_n are determined by the moments $M_{2k} = (\mathcal{O}, \mathcal{L}^{2k}\mathcal{O})_\phi$ of the modular two-point function via the Hankel determinant formula

$$b_n^2 \;=\; \frac{\det H_n \cdot \det H_{n-2}}{(\det H_{n-1})^2}, \qquad (H_k)_{ij} = M_{2(i+j)}.$$

On the truncation, by Remark 1 the modular flow agrees with the boost flow on $\le \Lambda_n$ modes, and the boost two-point function in the Bisognano–Wichmann KMS state has the universal hyperbolic decay

$$C(s) \;=\; (2\sinh(\pi s/\beta))^{-\Delta} \cdot (\text{spatial factors})$$

with $\beta = 2\pi$. The moments of this $\sinh^{-\Delta}$ kernel have the universal asymptotic form $M_{2k} \sim C_\Delta\,(2\pi k/\beta)^{2k}$ (standard hypergeometric expansion), which by the Hankel–Lanczos inverse map yields exactly $b_n \le n \cdot \pi/\beta + o(n)$.

Step 3 (combine). Substituting Step 2 into Step 1 gives (9). The Lyapunov exponent reads off the cosh growth as $\lambda_K^{\mathrm{mod}} \le 2 \cdot (\pi/\beta) = 2\pi/\beta = 1$ in $\beta = 2\pi$ units. □

Remark 2. *Theorem 1 is purely kinematic. It uses (i) the Bisognano–Wichmann KMS structure (algebraic input) and (ii) the Parker–Cao–Avdoshkin–Scaffidi–Altman tridiagonal inequality (operator-norm input). No chaos hypothesis, no holography, no large-N assumption is invoked.*

Remark 3. *For 2D CFT in the Bisognano–Wichmann KMS state, Caputa–Magán–Patramanis–Tonni [3] compute* $b_n^{\mathrm{CFT}} = n \cdot \pi/\beta$ *exactly, saturating* (8) *as an equality. Thus the bound is tight on the CFT class. The full type-II$_\infty$ saturation question (do all dressed type-II$_\infty$ algebras saturate?) is open – see §4.*

Corollary 2 (MSS bound recovery)**.** *For the truncation* $\mathcal{M}_R \otimes B(\mathcal{H}_n)$ *at any KMS inverse temperature* β*, the modular Lyapunov exponent satisfies* $\lambda_K^{\mathrm{mod}} \leq 2\pi/\beta$*, which is the MSS rate* (1)*.*

Proof. Replace $\beta = 2\pi$ by general β in the proof of Theorem 1; the hyperbolic moment kernel becomes $(2\sinh(\pi s/\beta))^{-\Delta}$ and the universal Lanczos slope becomes π/β, giving the MSS rate. □

4 Saturation as a conjecture: free-QFT counter-examples to the converse

The original ECI "modular shadow" draft [22] proposed the stronger statement that saturation $\lambda_K^{\mathrm{mod}} = 2\pi/\beta$ is a structural fingerprint of type-II$_\infty$. On audit (A11, 2026-05-05), this converse fails: there exist non-chaotic systems that saturate the kinematic ceiling. We collect three counter-examples and then state what survives as a conjecture.

Counter-example 1 (free-scalar lattice; Avdoshkin–Dymarsky 2019). Avdoshkin and Dymarsky [14] show that operator growth under imaginary-time evolution in 1D lattice models can attain exponential Lanczos-coefficient growth even in integrable regimes, with a 1D superexponential frequency suppression ensuring that "one-dimensional systems are special". This means linear b_n and exponential $\mathcal{C}_K$ are *not* signatures of chaos; they appear in integrable lattice systems as well.

Counter-example 2 (free & interacting massive scalar QFT; Camargo et al. 2023). Camargo, Jahnke, Kim, and Nishida [15] compute Krylov complexity in free and weakly interacting massive scalar field theories and find non-chaotic regimes where the Lanczos coefficients exhibit the same linear envelope as in maximally chaotic systems, with mass and UV cutoff modifying the prefactor but preserving the kinematic form. Their work refutes the reading that linear $b_n \to 2\pi/\beta$ implies type-II$_\infty$ MSS-saturating chaos.

Counter-example 3 (initial-operator dependence; Sreeram et al. 2025). Sreeram, Kannan, Modak, and Aravinda [16] (PRE 112 L032203) prove that Krylov complexity depends monotonically on the inverse participation ratio (IPR) of the initial operator in the Hamiltonian eigenbasis, and that this dependence persists in fully chaotic regimes. Saturation of λ_K^{mod} is therefore not a state- or operator-independent invariant of the algebra: it varies even within a fixed maximally-chaotic dynamics.

What survives as a conjecture. The strong claim of v1, "saturation $\Leftrightarrow$ type-II_∞", is refuted by counter-examples 1–3. What remains plausible is the weaker direction: type-II_∞ at $\beta = 2\pi$ is *sufficient* for the equality $b_n = n\pi/\beta$, even if not necessary.

Conjecture 1 (Modular saturation for dressed type-II_∞ algebras)**.** [CONJECTURE; v2 downgraded.] *Let $\mathcal{A}_{\mathrm{II}_\infty}$ be a type-II_∞ crossed product* (3) *arising from a QRF-dressing of a wedge or static-patch algebra, with the canonical $\beta = 2\pi$ KMS state. Then on a sequence of finite-rank truncations $\mathcal{M}_R \otimes B(\mathcal{H}_n)$, $n \to \infty$, the Lanczos coefficients of any few-mode operator $\mathcal{O}$ in a dense subalgebra satisfy*

$$b_n(\mathcal{O}) \xrightarrow[n\to\infty]{} n \cdot \frac{\pi}{\beta},$$

saturating the bound (8) *and yielding $\lambda_K^{\mathrm{mod}}(\mathcal{O}) = 2\pi/\beta$ in the large-truncation limit.*

Remark 4. *Conjecture 1 is consistent with the 2D-CFT computation of Caputa et al. [3] (Remark 3), with the SYK Krylov saturation of Parker et al. [2], and with the DSSYK / sine-dilaton holographic checks of Heller, Papalini, and Schuhmann [9]. We do not propose it as a theorem; the counter-examples above show that the converse fails, and even a one-sided proof of the sufficient direction is beyond current methods.*

Erratum: Faulkner–Speranza identification. The v1 draft of this work attributed to Faulkner and Speranza [6] the identification $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$. On rereading, FS24 derive the crossed-product generalised entropy and prove the GSL via Wall monotonicity; they do *not* explicitly identify S_{gen} with the modular Krylov complexity. The bridge $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$ is a working conjecture of the ECI programme, not a theorem of FS24. We thank the A11/A19 audit for catching this over-attribution.

5 Application: BEC analog Hawking radiation and experimental falsifiers

We now derive concrete observable consequences of Theorem 1 for the analog black-hole BEC platform of Kolobov, Golubkov, Muñoz de Nova, and Steinhauer [17] (Nat. Phys. 17, 362, 2021).

Remark 5 (Applicability caveat)**.** *Throughout this section we assume that the BEC exterior phonon algebra at finite healing length ξ admits a finite-rank type-II_∞ truncation in the sense of §2, following the crossed-product construction of Witten [4] and the Verchfolium argument for KMS states at T_H in the phononic ($k\xi \ll 1$) regime. This assumption is not proven for BEC systems; it is the operative hypothesis whose falsification is the goal of the experiments below.*

5.1 Hawking temperature and the predicted bound

Muñoz de Nova et al. [18] (Nature 569, 688, 2019) measured the analog Hawking temperature in the BEC sonic-horizon platform as

$$T_H \;=\; 0.35 \pm 0.10\,\mathrm{nK}, \tag{10}$$

implying the Bisognano–Wichmann KMS rate

$$\boxed{\Gamma_{\text{bound}} := \frac{2\pi k_B T_H}{\hbar} = (288 \pm 82)\,\text{s}^{-1}} \tag{11}$$

($k_B/\hbar = 1.31 \times 10^{11}\,\text{rad}\,\text{s}^{-1}\,\text{K}^{-1}$).

Under the assumption of Remark 5, Theorem 1 implies that any temporal correlator of exterior phonons must have its envelope decaying with rate $\leq \Gamma_{\text{bound}}$:

$$\Gamma_{\text{meas}} \leq \Gamma_{\text{bound}} = (288 \pm 82)\,\text{s}^{-1}, \tag{12}$$

where Γ_{meas} is the exponential-decay rate of any of: (i) the second-order coherence $g^{(2)}(\tau)$; (ii) the two-point density correlator $G^{(2)}(z_1, z_2, t)$ envelope; (iii) the Hawking-flux ramp-up rate.

5.2 Bound and saturation conjectures

The prediction (12) is a one-sided bound. It is convenient to split the experimental content into two forms, following the modular-shadow finite-rank theorem. Define $\lambda_L^{\text{sonic}} := 2\pi k_B T_H/\hbar = \Gamma_{\text{bound}}$.

Conjecture 2 (Bound form)**.** *For phonons near the sonic horizon of a BEC analog black hole at temperature T_H in the linear-dispersion regime ($k\xi \ll 1$), the temporal decay rate Γ_{meas} of the second-order coherence $g^{(2)}(\tau)$ satisfies*

$$\Gamma_{\text{meas}} \leq \lambda_L^{\text{sonic}} = \frac{2\pi\, k_B\, T_H}{\hbar}, \tag{13}$$

where T_H is measured independently from the Hawking correlation spectrum. This is the provable form: it follows from Theorem 1 under Remark 5, via Parker et al. [2] and the Bisognano–Wichmann modular-flow identification.

Conjecture 3 (Saturation form)**.** *Under the same assumptions, the bound is saturated:*

$$\Gamma_{\text{meas}} = \lambda_L^{\text{sonic}} = \frac{2\pi\, k_B\, T_H}{\hbar}. \tag{14}$$

Status. Conjecture 3 is an open working hypothesis. The saturation step requires the modular two-point function to take the universal hyperbolic Bisognano–Wichmann form $C(s) \propto (2\sinh(\pi s/\beta))^{-\Delta}$; this is automatic for genuine geometric/holographic flow but is *not* guaranteed in the BEC setting. The free-QFT and integrable counter-examples of §4 show that exponential Krylov growth saturating the bound can occur without chaos [14–16]. Hence saturation $\Rightarrow$ type-II$_\infty$ chaoticity is false in general, and Conjecture 3 cannot be deduced from Conjecture 2 alone.

Falsifier regimes. For Steinhauer 2019 parameters $T_H = 0.35 \pm 0.10\,\text{nK}$ the central value is $\lambda_L^{\text{sonic}} = 288\,\text{s}^{-1}$. Combining 1% statistical precision (at $N_{\text{pairs}} = 10^4$) with the dominant 29% systematic from T_H uncertainty:
The two conjectures are not equivalent: the bound form admits $\Gamma_{\text{meas}} < \lambda_L^{\text{sonic}}$ as compatible, while the saturation form demands equality.

Γ_{meas} regime	Conj. 2 (Bound)	Conj. 3 (Saturation)
$< 0.5\,\lambda_L^{\text{sonic}}$ ($\lesssim 144\,\text{s}^{-1}$)	consistent	refuted
$\in [0.8,\ 1.2]\,\lambda_L^{\text{sonic}}$	consistent	consistent
$> 1.5\,\lambda_L^{\text{sonic}}$ ($> 432\,\text{s}^{-1}$) at 5σ	**refuted** (Thm. 1 falls)	refuted

Table 1: Falsifier regimes for the BEC $g^{(2)}(\tau)$ decay rate. The bound falsifier $\Gamma_{\text{meas}} > 432\,\text{s}^{-1}$ at 5σ refutes Theorem 1 independently of any chaos or saturation hypothesis.

5.3 Signal-to-noise analysis

We present the signal-to-noise estimate for a dedicated measurement of the $g^{(2)}(\tau)$ decay rate. All numerical values below are consistent with `sympy` verification on Steinhauer 2019 parameters [18].

5.3.1 Phonon pair flux and integration time

The thermal characteristic frequency at the sonic horizon in the linear-dispersion regime is

$$f_{\text{phonon}} \;=\; \frac{k_B\,T_H}{h} \;=\; \frac{\kappa}{4\pi^2} \;\approx\; 7.3\,\text{Hz}, \tag{15}$$

where $\kappa = 2\pi k_B T_H/\hbar$ is the acoustic surface gravity. This is the rate of Hawking phonon pairs emitted per mode per unit time.

To resolve the exponential decay $g^{(2)}(\tau) \propto e^{-\Gamma\tau}$ to relative error $\delta\Gamma/\Gamma < 5\%$, Fisher-information analysis (see below) requires $N_{\text{pairs}} \geq 400$. For a standard run at $N_{\text{pairs}} = 10^4$:

$$t_{\text{total}} \;=\; \frac{N_{\text{pairs}}}{f_{\text{phonon}}} \;\approx\; 1371\,\text{s} \;\approx\; 23\,\text{min}. \tag{16}$$

This is a conservative single-mode estimate; a 1D condensate with $\mathcal{O}(L/\xi) \sim 10^3$ modes in parallel would reduce the required time proportionally.

Quantity	Value	Unit
T_H (Steinhauer 2019)	0.35 ± 0.10	nK
Γ_{bound} (ECI prediction)	287.9 ± 82.3	s^{-1}
Krylov saturation time $1/\Gamma_{\text{bound}}$	3.47	ms
Phonon emission rate f_{phonon}	7.29	Hz
BEC coherence time	~ 100	ms
E-foldings within coherence window	~ 29	—
N_{pairs} for 5% precision	400	pairs
Integration time (single-mode, $N = 10^4$)	23	min
Statistical precision at $N = 10^4$	1.0	%
Systematic limit (T_H uncertainty)	29	%

Table 2: Signal-to-noise summary for the BEC $g^{(2)}(\tau)$ measurement at the sonic horizon.

5.3.2 Fisher information and Cramér–Rao bound

For the single-parameter model $g^{(2)}(\tau) - 1 = A\, e^{-\Gamma\tau}$, the maximum-likelihood estimator (MLE) for Γ from N_{pairs} pair-detection events has Fisher information

$$\mathcal{I}(\Gamma) \;=\; \frac{N_{\text{pairs}}}{\Gamma^2}, \tag{17}$$

giving Cramér–Rao standard error

$$\sigma_\Gamma \;=\; \frac{\Gamma}{\sqrt{N_{\text{pairs}}}}. \tag{18}$$

For $N_{\text{pairs}} = 10^4$:

$$\sigma_\Gamma \;=\; \frac{287.9}{\sqrt{10^4}} \;=\; 2.88\,\text{s}^{-1}, \qquad \frac{\sigma_\Gamma}{\Gamma} \;=\; 1.0\%. \tag{19}$$

The statistical SNR is therefore $\text{SNR} = \sqrt{N_{\text{pairs}}} = 100\sigma$.

5.3.3 Wilks likelihood ratio test

To distinguish the ECI model (H_1: exponential decay at rate Γ) from the null (H_0: constant $g^{(2)}$), the Wilks likelihood-ratio statistic is

$$\Lambda_W \;=\; -2\log\frac{\mathcal{L}(H_0)}{\mathcal{L}(H_1)} \;\xrightarrow{N\to\infty}\; \chi^2_1 \quad \text{under } H_0. \tag{20}$$

Under H_1, the non-centrality parameter is $\lambda_{\text{nc}} = N_{\text{pairs}} = 10^4$, giving a formal Z-score of $\sqrt{10^4} = 100\sigma$. The 5σ threshold requires only $N_{\text{pairs}} \geq 25$, trivially satisfied.

5.3.4 Dominant systematic: T_H uncertainty

The dominant practical limitation is the $\pm 29\%$ uncertainty in $T_H = 0.35 \pm 0.10\,\text{nK}$, propagating linearly to Γ:

$$\frac{\delta\Gamma}{\Gamma} \;=\; \frac{\delta T_H}{T_H} \;\approx\; 29\%. \tag{21}$$

The 1-sigma band of the ECI prediction is $\Gamma \in [205.6,\, 370.2]\,\text{s}^{-1}$. A precision test of the scaling law $\lambda_L^{\text{sonic}} = 2\pi k_B T_H/\hbar$ requires T_H to be re-measured with $< 5\%$ accuracy.

5.4 Kolobov 2021 re-analysis as a bound falsifier

The Kolobov–Golubkov–Muñoz de Nova–Steinhauer 2021 dataset [17] measured the time evolution of $G^{(2)}(z_1, z_2, t)$ at six timepoints across 97,000 experimental repetitions (124-day acquisition). Re-analysis of the temporal envelope of $G^{(2)}(z_1, z_2, t)$ — which, by the Wiener–Khintchine relation, is the temporal cousin of $g^{(2)}(\tau)$ — can extract a 1σ upper bound on Γ_{meas} directly from published data, without new experimental runs.

Proposition 3 (BEC bound falsifier)**.** *If a re-analysis of Kolobov 2021 data yields* $\Gamma_{\text{meas}} > 1.5 \times \Gamma_{\text{bound}} = 432\,\text{s}^{-1}$ *at* 5σ*, then the type-*II_∞ *finite-rank scaffolding underlying Theorem 1 is falsified for the analog-BEC exterior phonon algebra (under the assumption of Remark 5).*

Interpretation of Kolobov re-analysis.

- $\Gamma_{\text{meas}} \leq 288\,\text{s}^{-1}$: consistent with Theorem 1. No new physics required; the bound follows from existing modular-flow theory.
- $\Gamma_{\text{meas}} \in [205, 370]\,\text{s}^{-1}$ (within the 1σ band of λ_L^{sonic}): non-trivial corroboration of both Conjectures 2 and 3.
- $\Gamma_{\text{meas}} > 432\,\text{s}^{-1}$ at 5σ: **bound falsifier triggered.** Theorem 1 and the type-II_∞ modular-shadow scaffolding fall *independently of any chaos hypothesis.*

This re-analysis route is preferable to a fresh dedicated measurement because: (a) it requires no new data acquisition; (b) it tests the strongest (BOUND) form of the prediction; (c) it is robust to the integrability counter-examples [14–16] that constrain the saturation form.

Robustness to integrability counter-examples. The bound (12) is the BOUND form, not the saturation form: even if the BEC phonon system is effectively integrable on the relevant timescales, the bound still holds under the type-II_∞ assumption. Only an observed *violation* refutes the kinematic prefactor.

Cross-platform redundancy. Independent platforms can cross-check (12): polariton-condensate analog black-hole mergers (Solnyshkov–Paquelier–Balmisse–Malpuech 2026, arXiv:2603.01664 [19]) and water-tank vortex QNM decay rates [21] provide independent substrates. A UV-finite volume-law entanglement negativity (Chandran–Fischer 2026, arXiv:2604.02075 [20]) likewise yields an entanglement-growth rate bounded by (12) via Theorem 1.

6 Discussion: comparison with Vardian (2026) and scope

Vardian's concurrent boundary-area-operator construction. Vardian [13] (arXiv:2602.02675, February 2026, single-author hep-th) reconstructs the QES area operator from boundary modular Lanczos coefficients in AdS/CFT, using operator-algebra quantum error correction. Specifically: from boundary modular dynamics, the spectrum of the boundary modular Hamiltonian is extracted via the Lanczos algorithm; its central component is identified with the area operator of the dual quantum extremal surface. Applications discussed include diagnosing island formation and the Page transition for evaporating black holes.

Comparison: complementary scopes. Vardian's work and ours overlap in their use of modular Lanczos coefficients on a type-II_∞-like algebra, but the scopes differ along three axes:

- **Holographic context.** Vardian works in AdS/CFT and uses holographic boundary–bulk duality explicitly; we work intrinsically in the finite-rank truncation without invoking AdS/CFT and our results apply to any wedge or static-patch algebra (de Sitter, Rindler, BEC analog horizon).

- **Object of study.** Vardian targets the area operator (a specific element of the modular Hamiltonian's spectrum); we target the kinematic Lyapunov ceiling on Krylov growth (an envelope statement on all operators).
- **Claim type.** Vardian provides a constructive boundary–bulk dictionary; we provide a one-sided inequality with concrete falsifier on a laboratory analog system.

The two works together suggest that modular Krylov techniques are becoming a standard tool for type-II_∞ algebras, with Vardian probing the holographic side and our bound theorem covering the non-holographic side (de Sitter, BEC, and generic truncations).

Scope of Theorem 1. Our theorem applies to any finite-rank truncation of a type-II_∞ crossed product at canonical KMS temperature. This covers:

1. Rindler wedge of a free QFT (where the bound is tight by Caputa et al. [3]);
2. BTZ exterior in AdS_3 (the holographic paradigm);
3. de Sitter static patch [5], with $\beta = 2\pi/H$;
4. BEC and other analog-Hawking systems with a sonic horizon (Kolobov 2021, Solnyshkov 2026);
5. the SYK / DSSYK / sine-dilaton family at large N [2, 9].

It does *not* apply directly to:

1. Type-III_1 algebras at finite N without a QRF dressing (no semifinite trace, modular flow ergodic);
2. Cosmological algebras (Bianchi IX, FRW) where the type classification is open;
3. Type-I or II_1 models without boost geometry (no Bisognano–Wichmann input).

Modular vs physical Lyapunov exponents. The bound (9) is on the modular Lyapunov exponent λ_K^{mod}, defined via modular flow. Translating to the physical Lyapunov exponent $\lambda_L^{\mathrm{OTOC}}$ requires the physical Hamiltonian to align with the modular Hamiltonian under the Connes–Rovelli [12] thermal-time hypothesis. For boost-symmetric wedges this alignment is automatic (Bisognano–Wichmann); for general dynamical settings it is an additional input.

7 Outlook

1. **Lifting the truncation.** Theorem 1 is stated on the finite-rank truncation $\mathcal{M}_R \otimes B(\mathcal{H}_n)$. A genuine type-$\mathrm{II}_\infty$ statement requires a uniform-in-n control of b_n and a passage to the inductive-limit algebra. The Hankel-determinant moment estimate of Step 2 in our proof sketch is uniform in n on the BW class, but the operator $\mathcal{O}$ must lie in a dense $*$-subalgebra of the type-II_∞ parent. We expect the lifted statement to hold; a complete proof is left for future work.

2. **de Sitter static patch.** CLPW [5] establish the type-II_∞ structure of the dS static patch with $\beta = 2\pi/H$ (Gibbons–Hawking temperature). By Theorem 1, the modular Lyapunov exponent of any few-mode operator on the dS static patch is bounded by H (the Hubble rate). A concrete observable (e.g. a primordial-tensor-mode decoherence rate) testing this bound would be of cosmological interest.

3. **Bianchi IX and homogeneous chaotic cosmologies.** The mixmaster Bianchi IX universe exhibits chaotic dynamics with a known modular structure on the kasner-segment wedge algebra. Whether this algebra is type-II_∞ and whether its modular Krylov exponent saturates (9) is open.

4. **Connes–Størmer entropy.** The Connes–Størmer entropy on type-II_1 reductions of our truncation provides an alternative diagnostic (see Connes–Rovelli [12]); whether it inherits the same kinematic ceiling is an open question.

5. **Bridge $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$.** The bridge between the FS24 generalised entropy and modular Krylov complexity remains a working conjecture of the ECI programme. A rigorous derivation would close the loop between Theorem 1 and the Faulkner–Speranza GSL.

6. **Relation to Vardian's area-operator reconstruction.** The boundary area operator of Vardian [13] sits in the central part of the modular Hamiltonian. Its Krylov-Lanczos signature should sit at a specific level of our truncation; making this precise would unify the two approaches.

Acknowledgments

This work was extracted and reworked from the ECI framework (v6.0.53, Zenodo record `10.5281/zenodo.20030684`). The split between the (provable) bound and the (conjectural) saturation, and the identification of the Avdoshkin–Dymarsky / Camargo et al. / Sreeram et al. counter-examples, owe to the A11 audit (2026-05-05). The discovery of Vardian [13] as concurrent independent work was made via the same audit, with arXiv ID 2602.02675 live-verified. Erratum on the Faulkner–Speranza over-attribution (§4) follows the A11 cross-check. Acknowledgements deferred to final version.

A Full proof of the modular Lyapunov bound on finite-rank type-II_∞ truncations

This appendix gives the full LMP-class proof of Theorem 1, expanding the three-step sketch of §3 into a publication-quality argument. The two main pillars are:

1. The universal tridiagonal-Lanczos cosh inequality of Parker, Cao, Avdoshkin, Scaffidi, and Altman [2] on the KMS–GNS inner product (algebraic, no chaos hypothesis).

2. The Bisognano–Wichmann Källén–Lehmann spectral decomposition of the modular two-point function on the truncation (axiomatic QFT, no holography).

The saturation criterion (Step 3) is a corollary of the moment-problem inversion.

A.1 Setup and definitions

Throughout this appendix, $\mathcal{H}$ is a separable Hilbert space, $\mathcal{B}(\mathcal{H})$ the bounded operators, $\mathcal{A}_{\mathrm{III}_1}$ the type-III_1 wedge algebra of a relativistic QFT in a Rindler wedge $\mathcal{W}$, and $|\Omega\rangle$ the cyclic and separating vacuum vector inducing the KMS state ϕ at Bisognano–Wichmann inverse temperature $\beta_{\mathrm{BW}} = 2\pi$.

A.1.1 Crossed-product algebra and modular Hamiltonian

Following Witten [4] and Chandrasekaran–Longo–Penington–Witten (CLPW) [5], the crossed product

$$\mathcal{A}_{\mathrm{II}_\infty} \;:=\; \mathcal{A}_{\mathrm{III}_1} \rtimes_{\sigma^\phi} \mathbb{R} \tag{22}$$

is type-II_∞, admits a faithful semifinite normal trace τ, and contains a self-adjoint generator $K_\phi = -\beta^{-1}\log\Delta_\phi$, where Δ_ϕ is the Tomita–Takesaki modular operator of ϕ. The Bisognano–Wichmann theorem [10] identifies K_ϕ on $\mathcal{A}_{\mathrm{III}_1}$ with $2\pi K_{\mathrm{boost}}$, the wedge boost generator restricted to wedge-localised observables. In the crossed product, K_ϕ extends to an unbounded self-adjoint operator on the larger Hilbert space and generates the modular flow

$$\sigma_s : \mathcal{A}_{\mathrm{II}_\infty} \to \mathcal{A}_{\mathrm{II}_\infty}, \qquad \sigma_s(\mathcal{O}) \;=\; e^{iK_\phi s}\,\mathcal{O}\,e^{-iK_\phi s}, \qquad s \in \mathbb{R}. \tag{23}$$

A.1.2 Finite-rank truncation

For each $\Lambda > 0$, let P_Λ be the spectral projector of K_ϕ onto the modular-energy band $|K_\phi| \le \Lambda$, and choose Λ_n such that $\dim(\operatorname{ran} P_{\Lambda_n}) = n < \infty$. Define the finite-rank truncation

$$\mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n) \;:=\; P_{\Lambda_n}\,\mathcal{A}_{\mathrm{II}_\infty}\,P_{\Lambda_n}. \tag{24}$$

This is a finite-dimensional von Neumann algebra (an $M_n(\mathbb{C})$-summand inside the type-II_∞ parent). It inherits the trace τ (now finite) and a faithful KMS state $\phi_n := \tau/\tau(P_{\Lambda_n})$. Crucially, K_ϕ commutes with P_{Λ_n}, so the truncated modular Hamiltonian $K_R := P_{\Lambda_n} K_\phi P_{\Lambda_n}$ is a bounded self-adjoint operator generating modular flow on the truncation, $\sigma_s^{(n)}(\mathcal{O}) = e^{iK_R s}\,\mathcal{O}\,e^{-iK_R s}$.

Lemma 4 (Bisognano–Wichmann persistence on the truncation)**.** *The KMS condition at $\beta = 2\pi$ holds on $\mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n)$ with state ϕ_n and modular generator K_R. Moreover, for any operator $\mathcal{O} \in \mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n)$ of modular energy support $\subset [-\Lambda_n, \Lambda_n]$, the modular two-point function*

$$C_{\mathcal{O}}(s) \;:=\; \phi_n\big(\mathcal{O}^*\,\sigma_s^{(n)}(\mathcal{O})\big)$$

agrees on $|s| \le s_n$ (with $s_n \to \infty$ as $\Lambda_n \to \infty$) with the boost two-point function on the wedge, i.e. with the analytic continuation of the Wightman two-point function at boost angle s.

Sketch. The KMS condition is preserved under spectral truncation because $\Delta_\phi P_{\Lambda_n} = P_{\Lambda_n}\Delta_\phi$ implies σ_s restricts to $\sigma_s^{(n)}$ on $\operatorname{ran}(P_{\Lambda_n})$. The Bisognano–Wichmann identification $K_\phi = 2\pi K_{\mathrm{boost}}$ descends to $K_R = 2\pi P_{\Lambda_n} K_{\mathrm{boost}} P_{\Lambda_n}$; the boost two-point function inherits this structure on the modular-energy band $\le \Lambda_n$. The cutoff horizon $s_n = O(\log\Lambda_n)$ controls when spectral truncation begins to deform the analytic two-point function. $\square$

A.1.3 Modular Krylov complexity

Equip $\mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n)$ with the KMS–GNS inner product

$$(A, B)_{\phi_n} \;:=\; \tau\big(\rho^{1/2} A^* \rho^{1/2} B\big) \;=\; \langle \Omega_n | A^* \sigma^{(n)}_{i\beta/2}(B) | \Omega_n \rangle, \tag{25}$$

where $\rho := e^{-\beta K_R}/Z$ is the truncated KMS density operator and $|\Omega_n\rangle$ the cyclic vector. The modular Liouvillian

$$\mathcal{L} := i[K_R, \cdot\,] \tag{26}$$

acts on the operator Hilbert space $(\mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n), (\cdot,\cdot)_{\phi_n})$ as a self-adjoint generator of $\sigma^{(n)}_s$. (Self-adjointness follows from KMS reflection symmetry $(A, [K_R, B])_{\phi_n} = ([K_R, A], B)^*_{\phi_n}$; we use the Hermiticity convention with i explicit so that $\mathcal{L} = \mathcal{L}^\dagger$.)

Given $\mathcal{O} \in \mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n)$ with $(\mathcal{O}, \mathcal{O})_{\phi_n} = 1$, the Lanczos algorithm produces an orthonormal basis $\{|\mathcal{O}_n\rangle\}_{n\geq 0}$ of the Krylov subspace $\overline{\mathrm{span}}\{|\mathcal{O}\rangle, \mathcal{L}|\mathcal{O}\rangle, \mathcal{L}^2|\mathcal{O}\rangle, \ldots\}$ satisfying the three-term recurrence

$$\mathcal{L}\,|\mathcal{O}_n\rangle \;=\; b_{n+1}\,|\mathcal{O}_{n+1}\rangle + b_n\,|\mathcal{O}_{n-1}\rangle, \qquad b_n > 0, \quad |\mathcal{O}_{-1}\rangle := 0, \tag{27}$$

with $a_n \equiv 0$ by KMS reflection symmetry of $\mathcal{L}$ on $(\cdot,\cdot)_{\phi_n}$. The modular Krylov complexity is

$$\mathcal{C}^{\mathrm{mod}}_K(s) \;:=\; \sum_{n\geq 0} n\,|\varphi_n(s)|^2, \qquad e^{i\mathcal{L}s}|\mathcal{O}\rangle \;=\; \sum_{n\geq 0} \varphi_n(s)\,|\mathcal{O}_n\rangle. \tag{28}$$

A.2 Full proof of Theorem 1

The proof has three steps: (i) the Parker–Cao–Avdoshkin–Scaffidi–Altman tridiagonal-Lanczos inequality on the KMS–GNS inner product (operator-algebraic, no chaos); (ii) the Bisognano–Wichmann moment estimate via the Hankel determinant (kinematic, no holography); (iii) the inverse-Lanczos saturation criterion (reduction of equality to the two-point function form, with free-QFT counter-examples precluding a chaos reading).

Step 1: Tridiagonal-Lanczos cosh inequality on the KMS–GNS inner product.

Let $\mathcal{O} \in \mathcal{M}_R \otimes \mathcal{B}(\mathcal{H}_n)$ with $(\mathcal{O}, \mathcal{O})_{\phi_n} = 1$. Let $\mathcal{O}(s) = e^{i\mathcal{L}s}\mathcal{O}$ and decompose $\mathcal{O}(s) = \sum_n \varphi_n(s)\,|\mathcal{O}_n\rangle$ in the Krylov basis (27). The amplitudes $\varphi_n(s)$ satisfy the discrete Schrödinger equation

$$i\,\partial_s \varphi_n(s) \;=\; b_{n+1}\,\varphi_{n+1}(s) + b_n\,\varphi_{n-1}(s), \qquad \varphi_n(0) = \delta_{n,0}. \tag{29}$$

Sub-step 1a (energy conservation). The norm $\sum_n |\varphi_n(s)|^2 = 1$ is conserved (unitarity of $e^{i\mathcal{L}s}$ on the KMS–GNS Hilbert space).

Sub-step 1b (operator norm of the truncated tridiagonal generator). For the truncated tridiagonal matrix $T^{(N)}$ with entries $T^{(N)}_{n,n+1} = T^{(N)}_{n+1,n} = b_{n+1}$ and zero elsewhere, the standard Gershgorin bound gives $\|T^{(N)}\|_{\mathrm{op}} \leq 2\max_{1\leq n\leq N} b_n$.

Sub-step 1c (Parker et al. [2] cosh envelope). Following Parker–Cao–Avdoshkin–Scaffidi–Altman [2] on a tridiagonal Hamiltonian with linearly bounded coefficients $b_n \leq \alpha n$, the modular Krylov complexity satisfies

$$\mathcal{C}^{\mathrm{mod}}_K(s) \;\leq\; A\,\cosh(2\alpha s) \tag{30}$$

for some constant A depending on b_1 but not on s. The proof in [2] maps the recurrence (29) to a chiral Dirac equation on a half-line and applies a propagation-speed (Lieb–Robinson) bound, yielding the cosh growth as the saturating profile.

Adapting to the modular setting: replace the physical Hamiltonian H by K_R, the physical Liouvillian $[H, \cdot\,]$ by the modular Liouvillian $\mathcal{L} = i[K_R, \cdot\,]$, and the Gibbs inner product $\mathrm{tr}(e^{-\beta H}A^*B)$ by the truncated KMS–GNS inner product (25). The proof of (30) in [2] uses only the tridiagonal structure of the recurrence, the Hermiticity of the generator, and the unitarity of evolution; none of these properties are altered by substituting $\mathcal{L}$ for the physical Liouvillian.

Therefore: *if* $b_n \le \alpha n$, then (30) gives $\lambda_K^{\mathrm{mod}}(\mathcal{O}) \le 2\alpha$. This reduces Theorem 1 to establishing $\alpha \le \pi/\beta$, which is Step 2.

Step 2: Bisognano–Wichmann moment bound and the universal π/β slope.

The modular Lanczos coefficients b_n^2 are determined by the moments

$$M_{2k} := (\mathcal{O}, \mathcal{L}^{2k}\mathcal{O})_{\phi_n} \;=\; (-1)^k C_{\mathcal{O}}^{(2k)}(0), \tag{31}$$

i.e. by the even derivatives of the modular two-point function $C_{\mathcal{O}}(s) := \phi_n(\mathcal{O}^*\sigma_s^{(n)}(\mathcal{O}))$ at $s = 0$. The Hankel–Lanczos relation reads

$$b_n^2 \;=\; \frac{D_n \cdot D_{n-2}}{(D_{n-1})^2}, \qquad D_k := \det H_k, \quad (H_k)_{ij} = M_{2(i+j)}, \;\; 0 \le i, j \le k. \tag{32}$$

This is the standard Stieltjes–Chebyshev moment-problem formula; see Viswanath–Müller, *The Recursion Method* (1994).

Sub-step 2a (Bisognano–Wichmann spectral form on the truncation). By Lemma 4, on $|s| \le s_n$ the modular two-point function agrees with the boost two-point function of a wedge-localised operator. By Bisognano–Wichmann [10], this function in the KMS state at $\beta = 2\pi$ admits a Källén–Lehmann spectral decomposition over scaling-dimension weights:

$$C_{\mathcal{O}}(s) \;=\; \int_0^\infty d\Delta\; \rho_{\mathcal{O}}(\Delta)\,\big(2\sinh(\pi s/\beta)\big)^{-2\Delta} + (\text{spatial form factor}), \tag{33}$$

where $\rho_{\mathcal{O}}(\Delta) \ge 0$ is the operator-dependent weight on conformal dimensions Δ. This is the analogue of the Källén–Lehmann representation under modular flow, well-known in the algebraic-QFT literature; see Haag [11] (Chapter V) and Borchers' modular reflection identity. For the truncation, the integral (33) is cut off at the maximal scaling dimension $\Delta_n = O(\Lambda_n)$ accessible to the modular-energy band.

Sub-step 2b (moments of a hyperbolic kernel: explicit Mellin saddle). For each spectral weight $(2\sinh(\pi s/\beta))^{-2\Delta}$, the Taylor expansion at $s = 0$ involves only even powers, and the moments

$$M_{2k}^{(\Delta)} := (-1)^k \frac{d^{2k}}{ds^{2k}} \big(2\sinh(\pi s/\beta)\big)^{-2\Delta}\Big|_{s=0}$$

admit the closed-form asymptotic

$$M_{2k}^{(\Delta)} \;\sim\; C_\Delta\,(2\pi k/\beta)^{2k}\,\big(1 + O(1/k)\big), \qquad k \to \infty, \tag{34}$$

with $C_\Delta > 0$ depending on Δ but not on k. We give the explicit derivation via the inverse Mellin transform. The BW two-point function for scaling dimension Δ is

$$\langle \mathcal{O}(s)\mathcal{O}(0)\rangle_\beta \;=\; \big(\sinh^2(\pi s/\beta)\big)^{-\Delta}. \tag{35}$$

Its Mellin transform $\widetilde{C}(u) := \int_0^\infty s^{u-1}(\sinh^2(\pi s/\beta))^{-\Delta}\,ds$, analytic in $0 < \mathrm{Re}\,u < 2\Delta$, is

$$\widetilde{C}(u) \;=\; \Big(\frac{\beta}{\pi}\Big)^{2u} \frac{\Gamma(u)\,\Gamma(\Delta - u)}{\Gamma(\Delta)}. \tag{36}$$

The $2k$-th moment is then

$$M_{2k}^{(\Delta)} \;=\; \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \widetilde{C}(u)\,(2k)!\,u^{-2k-1}\,e^{-2k\log(\pi/\beta)}\,du, \tag{37}$$

with $c \in (0,\Delta)$. The saddle is located at $u^* = k$ (from $d/du[\log\widetilde{C}(u) - (2k+1)\log u] = 0$ and the digamma approximation $\psi(u) \approx \log u$ for large u). Steepest descent with Stirling's approximation gives

$$M_{2k}^{(\Delta)} \;\sim\; C_\Delta\left(\frac{2\pi k}{\beta}\right)^{2k}\bigl(1+O(1/k)\bigr), \tag{38}$$

confirming (34); see Whittaker–Watson, *Modern Analysis* (1927) §13.6 for the Mellin saddle-point estimate. Alternatively, the spectral representation $(2\sinh(\pi s/\beta))^{-2\Delta} = \int_{-\infty}^{\infty} e^{i\omega s}\tilde{\rho}_\Delta(\omega)\,d\omega$ gives the same conclusion directly: $\tilde{\rho}_\Delta(\omega)$ is peaked near $|\omega| \sim 2\pi k/\beta$ for the $2k$-th moment, by Fourier analysis of the hyperbolic kernel.

Integrating against $\rho_{\mathcal{O}}(\Delta)$ gives

$$M_{2k} \;\le\; C_{\mathcal{O}}\,(2\pi k/\beta)^{2k}\,\bigl(1+O(1/k)\bigr), \qquad k\to\infty, \tag{39}$$

where $C_{\mathcal{O}} := \int_0^{\Delta_n} d\Delta\,\rho_{\mathcal{O}}(\Delta)\,C_\Delta < \infty$ on the truncation.

Sub-step 2c (Hankel–Lanczos inversion). Substituting (39) into the Hankel formula (32) and using the standard result (Akhiezer, *Classical Moment Problem*, 1965, §1.4) that for moments $M_{2k} = m^{2k}\,\mu_k$ with μ_k subexponential the corresponding Lanczos coefficients satisfy $b_n = m\cdot n + o(n)$, one obtains

$$b_n(\mathcal{O}) \;\le\; n\cdot\frac{\pi}{\beta} + o(n), \qquad n\to\infty, \tag{40}$$

which is (8). Positivity of the Hankel determinants is inherited from the positive-semidefinite spectral weight $\rho_{\mathcal{O}} \ge 0$.

Step 3: Saturation criterion and the universal operator-growth hypothesis (UOGH).

Equality $b_n(\mathcal{O}) = n\pi/\beta$ holds asymptotically if and only if (34) saturates, i.e. the spectral weight $\rho_{\mathcal{O}}$ is concentrated on a single dimension or yields equivalent moment growth. By the inverse-Lanczos statement of Parker et al. [2, Theorem 2], specifying the asymptotic slope $\alpha = \pi/\beta$ uniquely determines the long-time exponential decay as $C_{\mathcal{O}}(s) \sim e^{-(\pi/\beta)s\cdot 2\Delta}$, equivalent to

$$C_{\mathcal{O}}(s) \;=\; N_\Delta\,\bigl(2\sinh(\pi s/\beta)\bigr)^{-\Delta}, \tag{41}$$

the Bisognano–Wichmann form. This is the UOGH of Parker et al. [2], in its modular incarnation.

The chaos reading is refuted by free-QFT counter-examples (see §4 for full discussion): Avdoshkin–Dymarsky [14] show linear b_n in integrable lattice models; Camargo–Jahnke–Kim–Nishida [15] show the same kinematic form in free and weakly-interacting massive scalar QFTs; Sreeram–Kannan–Modak–Aravinda [16] show saturation is operator-dependent within maximally-chaotic dynamics.

Combining Steps 1, 2, 3.

Step 2 gives $b_n(\mathcal{O}) \le n\cdot\pi/\beta + o(n)$, so $\alpha := \pi/\beta$ is a valid linear bound. Step 1 then yields $\mathcal{C}_K^{\rm mod}(s) \le A\cosh(2\pi s/\beta)$, giving $\lambda_K^{\rm mod}(\mathcal{O}) \le 2\pi/\beta$. At $\beta = 2\pi$ this is the MSS rate $\lambda_K^{\rm mod} \le 1$. Step 3 characterises saturation (41) as a kinematic, not dynamical, statement, with free-QFT counter-examples precluding a chaos reading. □

A.3 Comparison with Vardian (2026)

Vardian [13] (arXiv:2602.02675, February 2026, single-author hep-th) presents a complementary construction in AdS/CFT. The central object is the boundary modular Hamiltonian's spectrum, extracted via the Lanczos algorithm and identified through operator-algebra quantum error correction (OAQEC) with the area operator of the dual quantum extremal surface (QES).

Our Theorem 1 and Vardian's construction share the modular-Lanczos technical machinery but address disjoint physical questions. Vardian operates in AdS/CFT and targets a specific spectral element (the area operator); we target the kinematic Lyapunov ceiling on Krylov growth for all operators in any finite-rank type-II_∞ truncation, including non-holographic settings (de Sitter, BEC). The two works together confirm that modular Krylov techniques are a productive tool for type-II_∞ algebras.

A.4 Open problems from the full proof

1. **Lifting to the full type-II_∞ algebra.** Theorem 1 is on the finite-rank truncation. Passing to the inductive-limit algebra requires uniform-in-n control of b_n for operators in a dense $*$-subalgebra of $\mathcal{A}_{\mathrm{II}_\infty}$.

2. **Saturation as a sufficient (not necessary) condition.** Conjecture 1 (§4) remains open.

3. **Bridge $S_{\mathrm{gen}} \leftrightarrow \mathcal{C}_K^{\mathrm{mod}}$.** The Faulkner–Speranza [6] generalised entropy and $\mathcal{C}_K^{\mathrm{mod}}$ are conjecturally related; rigorous statement is open.

4. **Higher-genus extension.** As confirmed by the A77 investigation (2026-05-05), the genus-$g \geq 2$ modular bootstrap lives in a different functor category (Riemann surface partition functions / Virasoro blocks) from the type-II_∞ operator-Hilbert Lanczos framework of this appendix: the three pillars of Step 1–3 have no off-the-shelf higher-genus analogs. Extension to $g \geq 2$ is a genuine open problem requiring new mathematical technology.

References

[1] J. Maldacena, S. H. Shenker, and D. Stanford, "A bound on chaos," *JHEP* **08** (2016) 106, arXiv:1503.01409 [hep-th].

[2] D. E. Parker, X. Cao, A. Avdoshkin, T. Scaffidi, and E. Altman, "A Universal Operator Growth Hypothesis," *Phys. Rev. X* **9** (2019) 041017, arXiv:1812.08657 [cond-mat.str-el].

[3] P. Caputa, J. M. Magán, D. Patramanis, and E. Tonni, "Krylov complexity of modular Hamiltonian evolution," *Phys. Rev. D* **109** (2024) 086004, arXiv:2306.14732 [hep-th].

[4] E. Witten, "Gravity and the crossed product," *JHEP* **10** (2022) 008, arXiv:2112.12828 [hep-th].

[5] V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, "An algebra of observables for de Sitter space," *JHEP* **02** (2023) 082, arXiv:2206.10780 [hep-th].

[6] T. Faulkner and A. J. Speranza, "Gravitational algebras and the generalized second law," *JHEP* **11** (2024) 099, arXiv:2405.00847 [hep-th].

[7] J. De Vuyst, S. Eccles, P. A. Höhn, and J. Kirklin, "Crossed products and quantum reference frames: on the observer-dependence of gravitational entropy," *JHEP* **07** (2025) 063, arXiv:2412.15502 [hep-th].

[8] J. De Vuyst, S. Eccles, P. A. Höhn, and J. Kirklin, "Gravitational entropy is observer-dependent," *JHEP* **07** (2025) 146, arXiv:2405.00114 [hep-th].

[9] M. P. Heller, A. Papalini, and T. Schuhmann, "Krylov spread complexity as holographic complexity beyond JT gravity," *Phys. Rev. Lett.* **135** (2025) 151602, arXiv:2412.17785 [hep-th].

[10] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **17** (1976) 303.

[11] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed., Springer (1996).

[12] A. Connes and C. Rovelli, "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories," *Class. Quantum Grav.* **11** (1994) 2899–2917.

[13] N. Vardian, "Modular Krylov Complexity as a Boundary Probe of Area Operator and Entanglement Islands," arXiv:2602.02675 [hep-th] (February 2026).

[14] A. Avdoshkin and A. Dymarsky, "Euclidean operator growth and quantum chaos," *Phys. Rev. Research* **2** (2020) 043234, arXiv:1911.09672 [cond-mat.stat-mech].

[15] H. A. Camargo, V. Jahnke, K.-Y. Kim, and M. Nishida, "Krylov Complexity in Free and Interacting Scalar Field Theories with Bounded Power Spectrum," *JHEP* **05** (2023) 226, arXiv:2212.14702 [hep-th].

[16] Sreeram PG, J. Bharathi Kannan, R. Modak, and S. Aravinda, "Dependence of Krylov complexity on the initial operator and state," *Phys. Rev. E* **112** (2025) L032203, arXiv:2503.03400 [quant-ph].

[17] V. I. Kolobov, K. Golubkov, J. R. Muñoz de Nova, and J. Steinhauer, "Observation of stationary spontaneous Hawking radiation and the time evolution of an analogue black hole," *Nat. Phys.* **17** (2021) 362–367, arXiv:1910.09363 [gr-qc].

[18] J. R. Muñoz de Nova, K. Golubkov, V. I. Kolobov, and J. Steinhauer, "Observation of thermal Hawking radiation at the Hawking temperature in an analogue black hole," *Nature* **569** (2019) 688–691, arXiv:1809.00913 [gr-qc].

[19] D. D. Solnyshkov, I. Paquelier, A. Balmisse, and G. Malpuech, "Polariton-condensate analog black-hole merger" (working title), arXiv:2603.01664 (2026).

[20] A. Chandran and U. R. Fischer, "UV-finite volume-law entanglement negativity" (working title), arXiv:2604.02075 (2026).

[21] T. Torres, S. Patrick, A. Coutant, M. Richartz, E. W. Tedford, and S. Weinfurtner, "Rotational superradiant scattering in a vortex flow," *Nat. Phys.* **13** (2017) 833–836.

[22] [ECI author(s)], "Entanglement–Complexity–Information: a phenomenological architecture for quantum gravity phenomenology (v6.0.53)," Zenodo record `10.5281/zenodo.20030684` (2026).

[23] M. De Clerck, S. A. Hartnoll, and J. E. Santos, "Mixmaster chaos in an AdS black hole interior," *JHEP* **07** (2024) 202, arXiv:2312.11622 [hep-th].

[24] M. Requardt, "The Role of Type II_∞ v.Neumann Algebras and their Tensor Structure in Quantum Gravity," arXiv:2501.06009 [gr-qc] (2025).

[25] M. Motaharfar, M. R. Siebersma, and P. Singh, "Krylov Complexity in Canonical Quantum Cosmology," arXiv:2511.17711 (2025).

[26] A. J. Speranza, "An intrinsic cosmological observer," arXiv:2504.07630 [hep-th] (2025).